Ramya Yalamarthi

AI/ML Engineer
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AI/ML Engineer with 5+ years of experience and an M.S. in Information Technology, designing production-grade generative AI analytics and scalable data platforms that convert natural-language business questions into accurate Redshift SQL, cutting analyst query turnaround by 60%. Builds resilient RAG and self-healing NL-to-SQL systems with authorization and error-recovery safeguards, improving execution accuracy and reliability. Seeking to apply large-scale ML and cloud engineering to drive analytic automation and faster business insights.

Education

Master of Science, Information Technology, Jan 2024 – Aug 2025
Florida Atlantic University, Florida

B.Tech in Computer Science, Aug 2015 – May 2019
Jawaharlal Nehru Technological University, Hyderabad

Employment history

AI/ML Engineer, Nov 2025 – Present
Quest Global, Seattle, USA

- Architected a production multi-agent GenAI analytics system on AWS Bedrock AgentCore using the Strands Agents framework, orchestrating agents with Claude Opus (supervision) and Claude Sonnet (SQL generation) to convert natural-language business questions into executable Redshift SQL, reducing analyst query turnaround by 60%.
- Designed a retrieval-augmented generation (RAG) pipeline that dynamically retrieves database schema, business glossary, and 50+ KPI formula definitions (RPI, CPI, Margin, Match Rate) from DynamoDB and Amazon S3 to ground SQL generation, improving SQL execution accuracy by 40% and eliminating hallucinated column references.
- Built a self-healing Natural Language-to-SQL engine on Redshift Serverless with AWS Lambda and the Redshift Data API, featuring auto-retry on query errors, a custom SQL authorization layer (blocking DML/DDL, read-only enforcement, prompt-injection protection), and a code-interpreter fallback (pandas/numpy/scipy) for result sets >20 rows, improving analytical insight accuracy by 35%.
- Engineered a Recharts-compatible visualization agent powered by Claude Haiku that auto-generates chart configurations, KPIs, and narrative insights from query results, supporting 13 chart types and enabling zero-click BI dashboards for non-technical stakeholders.
- Implemented extended thinking with a 4K-token budget and context-window summarization (30% compression via a dedicated Haiku model), enabling long multi-turn analytics sessions while maintaining sub-second first-token latency through Server-Sent Events (SSE) streaming.
- Built scalable MLOps deployment pipelines using AWS CDK, CodeBuild, Lambda, ECR, and S3, automating ARM64 Docker image builds and Bedrock AgentCore Runtime deployments via Lambda custom resources, cutting release turnaround from hours to 15 minutes and reducing DevOps overhead by 60%.
- Implemented enterprise authentication using Azure AD OAuth (JWT validation) and AWS SigV4 signing, combined with role-based access control (RBAC) via group membership to enforce row- and schema-level data permissions across Redshift.
- Instrumented end-to-end observability with CloudWatch and Langfuse, capturing distributed traces, per-tool token/cost/latency metrics, and session correlation IDs across the multi-agent pipeline, enabling rapid debugging of non-deterministic agent behavior in production.

Sr. Product Engineer, Jul 2022 – Dec 2023
Simplilearn, Bangalore, India

- Designed self-serve Tableau and Power BI dashboards on transactional and engagement event data, tracking 20+ product KPIs (course enrollments, completion rates, learner engagement, instructor performance) across 5+ customer segments, eliminating 40% of ad-hoc analytics requests.
- Prototyped Simplilearn's first GenAI pilot using OpenAI GPT-3.5/GPT-4 with retrieval-augmented generation (RAG) over course content, enabling instant conversational help scoped to enrolled modules and contributing to repeat course enrollments.
- Launched a learner engagement analytics dashboard on Tableau tracking time-per-module, return frequency, session depth, and drop-off points, informing instructional design decisions that reduced mid-course drop-off by 35%.
- Built automated ingestion pipelines on AWS that integrated structured and unstructured data from enrollment forms, learner demographics, and platform utilization feeds into Redshift, enabling unified reporting and downstream analytics.
- Standardized metric definitions and democratized learner data access across product, sales, and operations teams, cutting stakeholder reporting time by 30% and enabling consistent KPI reporting organization-wide.

Software Engineer, Jan 2021 – May 2022

Kristal.AI, Bangalore, India

- Built and shipped investor-facing features on a digital private wealth platform serving clients across Singapore, Hong Kong, India, and UAE, using React, Node.js, and REST APIs to deliver a unified experience across four regulatory jurisdictions.
- Developed microservices powering portfolio management and investment execution workflows, handling multi-currency transactions and improving reliability of order processing across heterogeneous market environments.
- Implemented real-time portfolio tracking dashboards enabling investors to monitor performance across equities, ETFs, and alternative asset classes, with live valuation and allocation views.
- Analyzed user engagement on the investment platform using Python, SQL, and PySpark, uncovering that fewer than 5% of time-sensitive market alerts were opened during peak trading hours — insights that informed product decisions around alert timing, delivery channel, and notification UX.

Software Developer, Jun 2019 – Nov 2020

Accenture, Hyderabad, India

- Developed and optimized Ab Initio data-processing pipelines and SAP ABAP interfaces across ECC and CRM systems, using Oracle SQL for complex transformations, reducing batch processing time by 30%.
- Automated configuration sync workflows between ECC and MDG, reducing incident volume by 40% and improving platform stability.
- Refactored legacy pipelines using CI/CD with GitHub Actions and automated testing, improving pipeline reliability and reducing processing time by 45%.
- Implemented data quality and governance controls across SAP master data pipelines, including null checks, uniqueness validation, data drift detection, and secure handling practices, reducing downstream data incidents and improving audit readiness.

Certifications

AWS Certified Developer - Associate , Mar 2026

AWS Certified Generative AI Developer - Professional, Apr 2026

Technical Skills

Programming & Frameworks: Python (FastAPI), SQL, PySpark, PyTorch, Hugging Face Transformers, LangChain, Strands Agents, NumPy, Pandas

GenAI & LLMs: Large Language Models (LLMs), Multi-agent Systems, AWS Bedrock AgentCore, Retrieval-Augmented Generation (RAG), NL-to-SQL, Prompt Engineering, Extended Thinking, NLP (NER, Summarization, Sentiment Analysis), Generative AI

MLOps & DevOps: AWS CDK, CodeBuild, Lambda, ECR, Docker, Terraform, GitHub Actions, Apache Airflow, CI/CD Pipelines, Experiment Tracking, Model Registry, Git

Cloud Platforms & Big Data: AWS (Bedrock, S3, EC2, Lambda, Redshift, Redshift Serverless, DynamoDB), Azure (Databricks, Data Factory, Synapse, Event Hubs, ML), Snowflake, Apache Spark, Kafka, Event-Driven Architectures, REST APIs

Observability & Responsible AI: Langfuse, AWS CloudWatch, Distributed Tracing, Token/Cost/Latency Metrics, Session Correlation, Model Evaluation, A/B Testing

Data Engineering & Streaming: ETL/ELT Pipelines, Real-time Data Ingestion, Server-Sent Events (SSE), Redshift Data API, Apache Spark Streaming

Visualization & Reporting: Recharts, Tableau, Jupyter Notebooks, Matplotlib, Seaborn, Plotly

Workflow & Methodologies: Agile, Scrum, SDLC